

Computational Prediction of Drug-Induced Heart Damage: ADMET-AI vs SwissADME Comparison Analysis of 25 Cardiac RODEO Drugs

Cardiac RODEO Project

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Abstract

This report compares two computational approaches for predicting drug-induced heart damage: ADMET-AI (41 ADMET properties) and SwissADME (43 physicochemical properties). We replicate published DICTrank benchmark results, evaluate predictions on 25 Cardiac RODEO drugs, and train new models using Leave-One-Out Cross-Validation (LOOCV) and scaffold-balanced CV.

1 DICTrank Model: Training and 25-Drug Inference

Part 1: DICTrank model replication. We retrain the published DICTrank models on the DICTrank dataset (555 drugs: 293 most DICT concern, 262 no concern) using 10-fold scaffold-balanced cross-validation to confirm benchmark performance.

Part 2: 25-drug inference. We then generate features from the 25 Cardiac RODEO SMILES and pass them through the trained DICTrank models to obtain DICT concern probabilities (no retraining on the 25-drug set).

1.1 Process Summary

- **Training data:** DICTrank 555-drug dataset (most vs no concern).
- **Models:** GradientBoostingClassifier (n_estimators=500, max_depth=12, learning_rate=0.1, random_state=0).
- **Validation:** 10-fold scaffold-balanced CV; fold test predictions concatenated for ROC/AUC and confusion matrices.
- **Decision rule:** probability ≥ 0.5 for DICT concern.
- **25-drug inference:** ADMET-AI features computed for 25 drugs; SwissADME features available for 23 drugs, then passed through the trained DICTrank models (no retraining).
- **SwissADME coverage:*** Dactinomycin and Plicamycin are excluded due to size; SwissADME columns are shown as “-”.

Table 1: DICTrank Training Results (10-Fold CV)

Model	ROC AUC	PR AUC	Accuracy
ADMET-AI	0.72	0.75	0.64
SwissADME	0.66	0.68	0.60

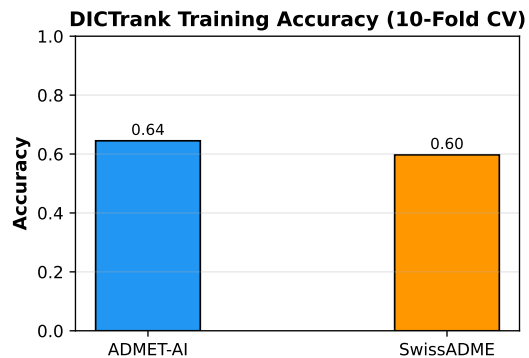
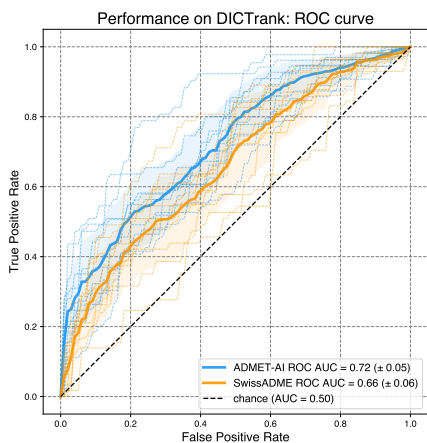


Figure 1: DICTrank training ROC and accuracy (10-fold CV on 555 drugs).

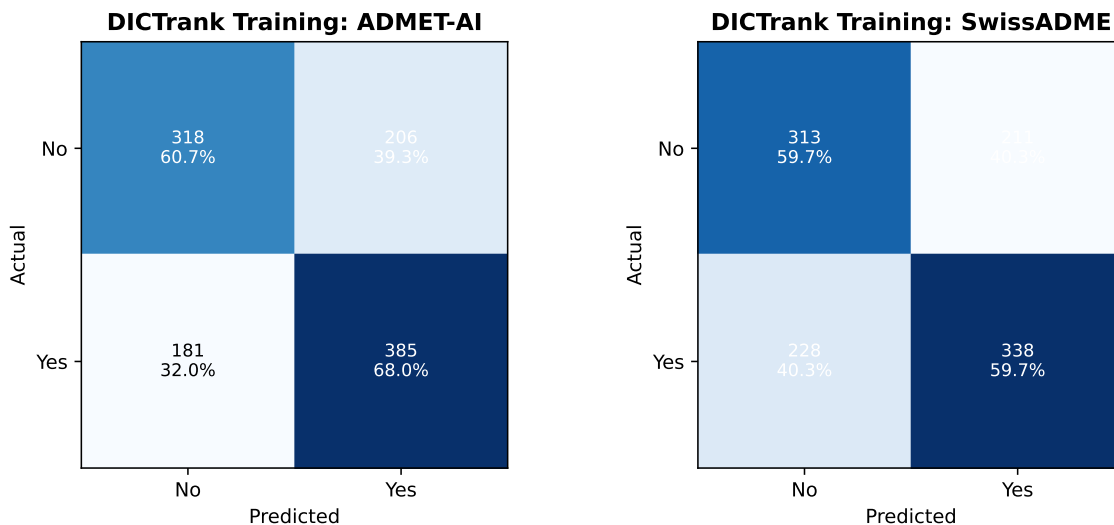


Figure 2: DICTrank training confusion matrices (10-fold CV on 555 drugs).

1.2 Cardiac RODEO Drug Dataset

Our dataset contains 25 drugs with experimentally determined cardiac outcomes:

- Heart Damage Positive: 20/25 drugs (80)
- Heart Damage Negative: 5/25 drugs (20)

Table 2: Drug Database with SMILES and Sources

Drug	CID	MW	SMILES
Amiodarone	2157	645.3	<chem>CCCCc1oc2c(c1C(=O)c1cc(I)c(c(c1)I)OCCN(CC)CC)cccc2</chem>

Drug	CID	MW	SMILES
Bortezomib	387447	384.2	<chem>CC(C[C@@H](B(O)O)NC(=O)[C@@H](NC(=O)C1CNCN1)Cc1cccc1)C</chem>
Chlorpromazine	2726	318.9	<chem>CN(CCCN1c2cccc2Sc2c1cc(C1)cc2)C</chem>
Cobimetinib	16222096	531.3	<chem>Ic1ccc(c(c1)F)Nc1c(ccc(c1F)F)C(=O)N1CC(C1)(O)[C@@H]1CCCCN1</chem>
Dactinomycin	457193	1255.4	<chem>C[C@@H]1[C@@H](C(=O)N[C@@H](C(=O)N2CCC[C@H]2C(=O)N(CC(=O)N([C@H](C(=O)O1)C(C)C)C)C(C)C)NC(=O)C3=C4C(=C(C=C3)C)OC5=C(C(=O)C(=C(C5=N4)C(=O)N[C@H]6[C@H](OC(=O)[C@@H](N(C(=O)CN(C(=O)[C@@H]7CCCN7C(=O)[C@H](NC6=O)C(C)C)C)C(C)C)N)C</chem>
Daunorubicin	30323	527.5	<chem>C0c1cccc2c1C(=O)c1c(C2=O)c(O)c2c(c1O)[C@@H](O[C@H]1C[C@H](N)[C@@H]([C@H](O1)C)O)C[C@](C2)(O)C(=O)C</chem>
Doxorubicin	31703	543.5	<chem>OCC(=O)[C@@]1(O)C[C@H](O[C@H]2C[C@H](N)[C@@H]([C@@H](O2)C)O)c2c(C1)c(O)c1c(c2O)C(=O)c2c(C1=O)cccc2OC</chem>
Epirubicin	41867	543.5	<chem>OCC(=O)[C@@]1(O)C[C@H](O[C@H]2C[C@H](N)[C@H]([C@@H](O2)C)O)c2c(C1)c(O)c1c(c2O)C(=O)c2c(C1=O)cccc2OC</chem>
Erlotinib	176870	393.4	<chem>COCCOc1cc2c(ncnc2cc1OCCOC)Nc1cccc(c1)C#C</chem>
Etomoxir	9840324	326.8	<chem>CCOC(=O)[C@]1(CCCCCOc2ccc(cc2)Cl)OC1</chem>
Gemcitabine	60750	263.2	<chem>OC[C@H]1O[C@H](C([C@@H]1O)(F)F)n1ccc(nc1=O)N</chem>
Ibrutinib	24821094	440.5	<chem>C=CC(=O)N1CCC[C@H](C1)n1nc(c2c1ncnc2N)c1ccc(cc1)Oc1cccc1</chem>
Ibuprofen	3672	206.3	<chem>CC(Cc1ccc(cc1)C(C(=O)O)C)C</chem>
Isoproterenol	3779	211.3	<chem>CC(NCC(c1ccc(c(c1)O)O)O)C</chem>
Mexiletine	4178	179.3	<chem>CC(COc1c(C)cccc1C)N</chem>
Nifedipine	4485	346.3	<chem>COC(=O)C1=C(C)NC(=C(C1c1cccc1[N+](=O)[O-])C(=O)OC)C</chem>
Panobinostat	6918837	349.4	<chem>ONC(=O)/C=C/c1ccc(cc1)CNCCc1c(C)[nH]c2c1cccc2</chem>
Plicamycin	163659	1085.1	<chem>C[C@@H]1[C@H]([C@@H](C[C@@H](O1)O[C@@H]2C[C@@H](O[C@@H]([C@H]2O)C)OC3=CC4=CC5=C(C(=O)[C@H]([C@@H](C5)[C@@H](C(=O)[C@H]([C@@H](C)O)O)OC)O[C@H]6C[C@H]([C@@H]([C@H](O6)C)O)O[C@H]7C[C@H]([C@H]([C@H](O7)C)O)O[C@H]8C[C@]([C@@H]([C@H](O8)C)O)(C)O)C(=C4C(=C3C)O)O)O)O</chem>
Rosiglitazone	77999	357.4	<chem>O=C1NC(=O)C(S1)Cc1ccc(cc1)OCCN(c1ccccn1)C</chem>
Sotalol	5253	272.4	<chem>CC(NCC(c1ccc(cc1)NS(=O)(=O)C)O)C</chem>
Sunitinib	5329102	398.5	<chem>CCN(CCN(C(=O)c1c(C)[nH]c(c1C)/C=C/1\{ }C(=O)Nc2c1cc(F)cc2)CC</chem>

Drug	CID	MW	SMILES
Vandetanib	3081361	475.4	<chem>C0c1cc2c(ncnc2cc1OCC1CCN(CC1)C)Nc1ccc(cc1F)Br</chem>
Vincristine	5978	825.0	<chem>O=CN1c2cc(OC)c(cc2[C@]23[C@@H]1[C@@](O)(C(=O)OC)[C@H](OC(=O)C)[C@]1([C@@H]3N(CC2)CC=C1)CC)[C@]1(C[C@@H]2CN(CCC3c1[nH]c1c3cccc1)C[C@](C2)(O)CC)C(=O)OC</chem>
Vioxx	5090	314.4	<chem>O=C1OCC(=C1c1cccc1)c1ccc(cc1)S(=O)(=O)C</chem>
Vorinostat	5311	264.3	<chem>ONC(=O)CCCCCCC(=O)Nc1cccc1</chem>

All drugs sourced from PubChem (<https://pubchem.ncbi.nlm.nih.gov>).

1.3 DICTrunk Inference on 25 Drugs

Table 3: DICTrunk Model Performance (ADMET-AI: 25 drugs, SwissADME: 23 drugs)

Model	Accuracy	ROC AUC	TP	FN	TN	FP
ADMET-AI	0.56	0.45	12	8	2	3
SwissADME	0.57	0.60	11	9	2	1

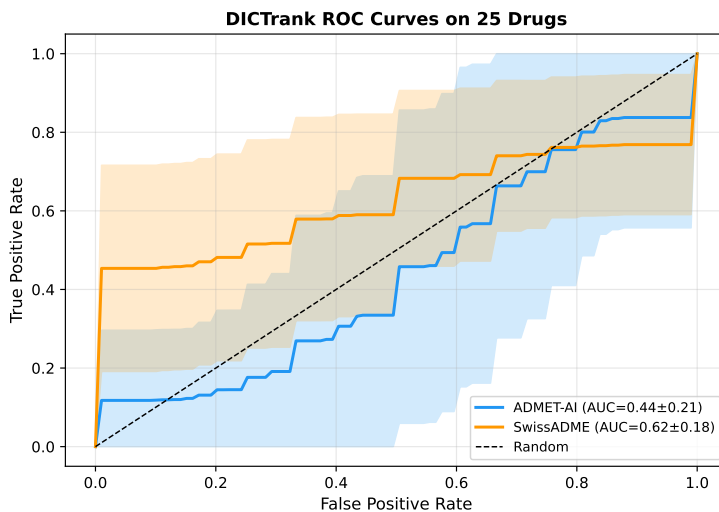
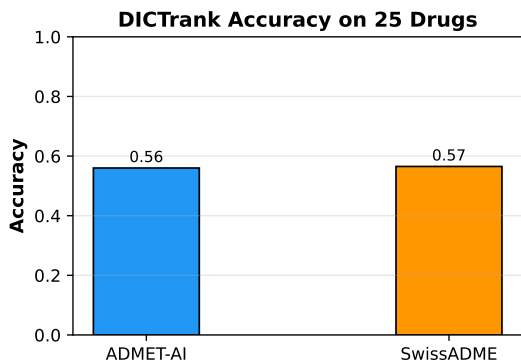


Figure 3: DICTrunk accuracy and ROC (ADMET-AI: 25 drugs, SwissADME: 23 drugs).

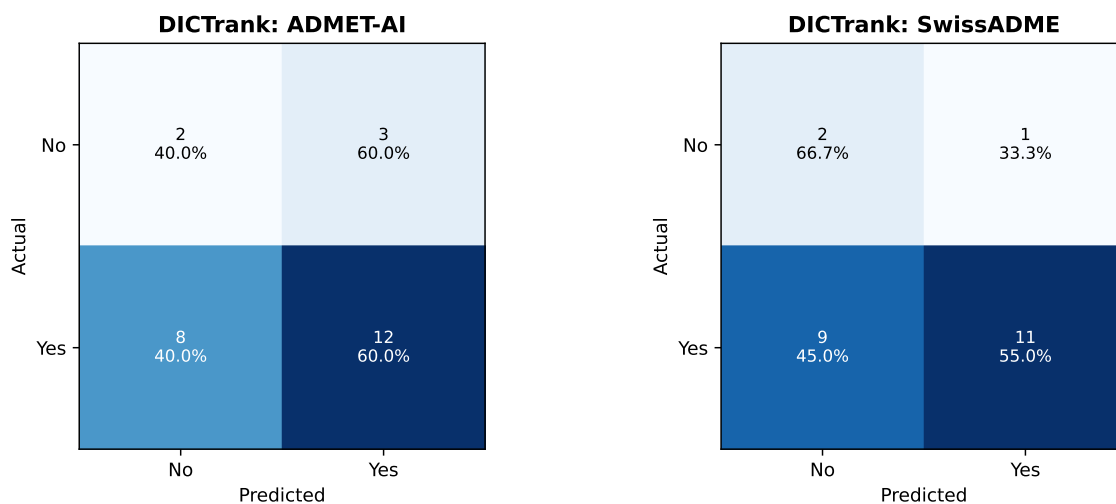


Figure 4: Confusion matrices for DICTrank models (ADMET-AI: 25 drugs, SwissADME: 23 drugs).

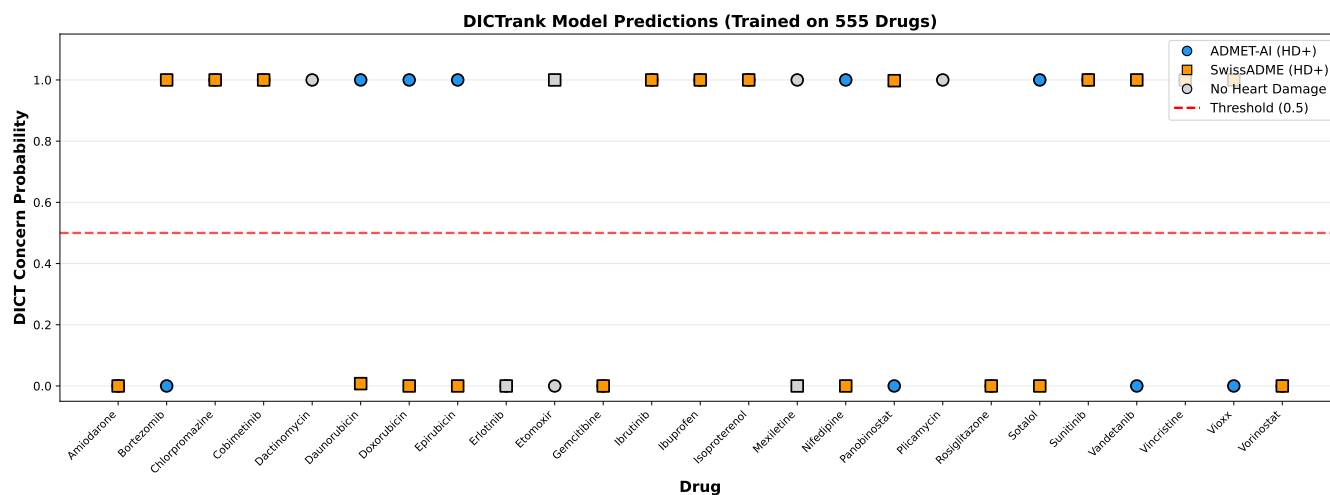


Figure 5: DICTrank predictions for each drug. Circles = ADMET-AI, Squares = SwissADME*.

*SwissADME unavailable for Dactinomycin and Plicamycin due to size; SwissADME columns shown as “_”.

Table 4: Individual Drug Predictions (DICTrank Models)

Drug	True HD	ADMET Prob	Pred	Swiss Prob	Pred
Amiodarone	Yes	0.000	Low	0.000	Low
Bortezomib	Yes	0.000	Low	1.000	High
Chlorpromazine	Yes	1.000	High	1.000	High
Cobimetinib	Yes	1.000	High	1.000	High
Dactinomycin	No	1.000	High	—	—
Daunorubicin	Yes	1.000	High	0.008	Low
Doxorubicin	Yes	1.000	High	0.000	Low
Epirubicin	Yes	1.000	High	0.000	Low
Erlotinib	No	0.000	Low	0.000	Low

Drug	True HD	ADMET Prob	Pred	Swiss Prob	Pred
Etomoxir	No	0.000	Low	1.000	High
Gemcitabine	Yes	0.000	Low	0.000	Low
Ibrutinib	Yes	1.000	High	1.000	High
Ibuprofen	Yes	1.000	High	1.000	High
Isoproterenol	Yes	1.000	High	1.000	High
Mexiletine	No	1.000	High	0.000	Low
Nifedipine	Yes	1.000	High	0.000	Low
Panobinostat	Yes	0.000	Low	0.998	High
Plicamycin	No	1.000	High	–	–
Rosiglitazone	Yes	0.000	Low	0.000	Low
Sotalol	Yes	1.000	High	0.000	Low
Sunitinib	Yes	1.000	High	1.000	High
Vandetanib	Yes	0.000	Low	1.000	High
Vincristine	Yes	1.000	High	1.000	High
Vioxx	Yes	0.000	Low	1.000	High
Vorinostat	Yes	0.000	Low	0.000	Low

2 LOOCV Models (25 Drugs)

We retrain ADMET-AI and SwissADME models directly on the 25 Cardiac RODEO drugs.

Table 5: LOOCV Model Performance

Model	Accuracy	ROC AUC	TP	FN	TN	FP
ADMET-AI	0.60	0.39	15	5	0	5
SwissADME	0.83	0.00	19	1	0	3

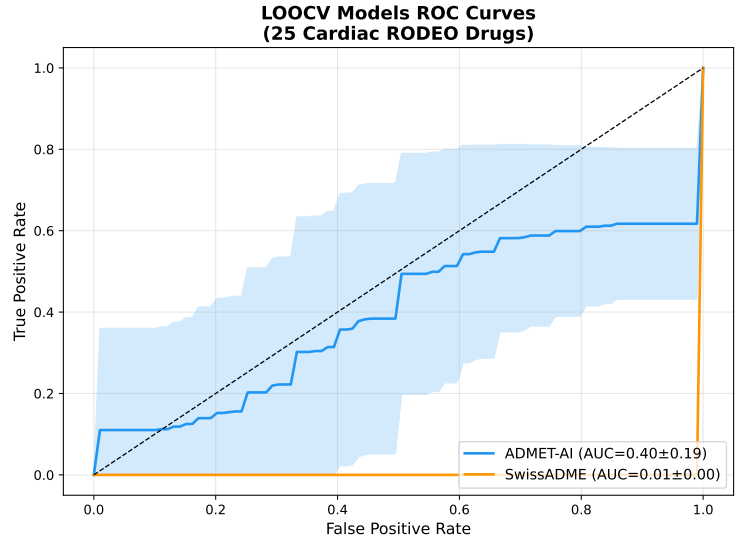
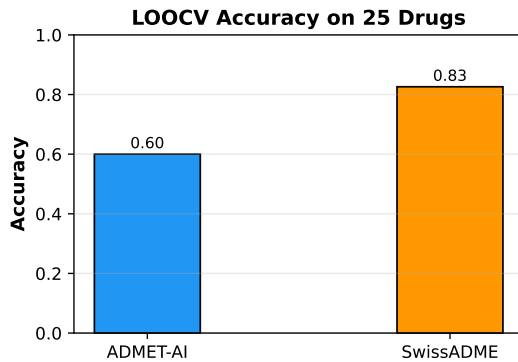


Figure 6: LOOCV accuracy and ROC on 25 drugs.

LOOCV Model Confusion Matrices

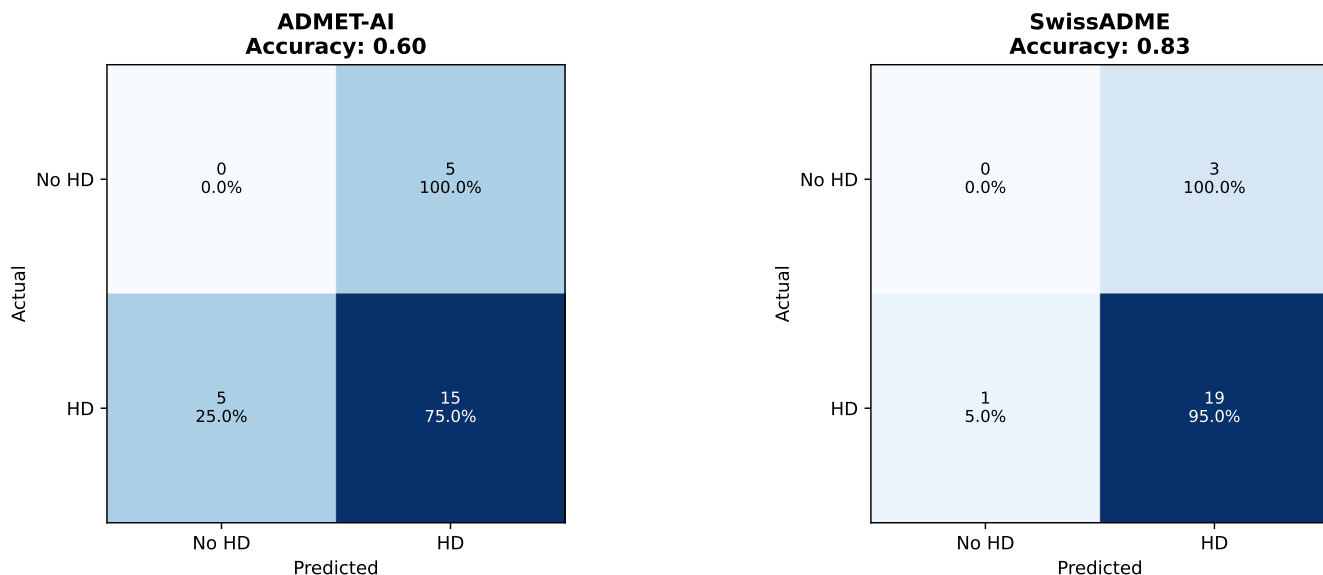


Figure 7: LOOCV confusion matrices (percent correct per class).

Note: Accuracy shown in the confusion-matrix titles is fold-averaged across LOOCV splits, while the matrix aggregates pooled predictions, so the diagonal may not match the title value.

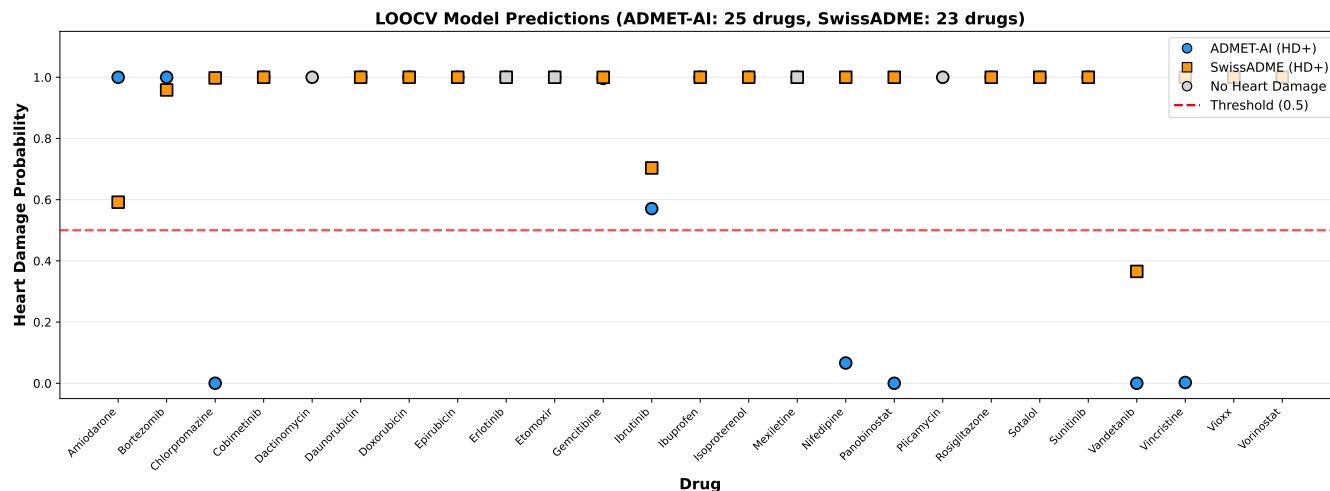


Figure 8: LOOCV predictions for each drug. Circles = ADMET-AI, Squares = SwissADME.

3 Scaffold-Balanced CV Models (25 Drugs)

Table 6: Scaffold-Balanced CV Model Performance (25 Drugs)

Model	Accuracy	ROC AUC (mean)	TP	FN	TN	FP
ADMET-AI	0.54	0.44	27	11	0	12
SwissADME	0.80	0.38	40	20	15	20

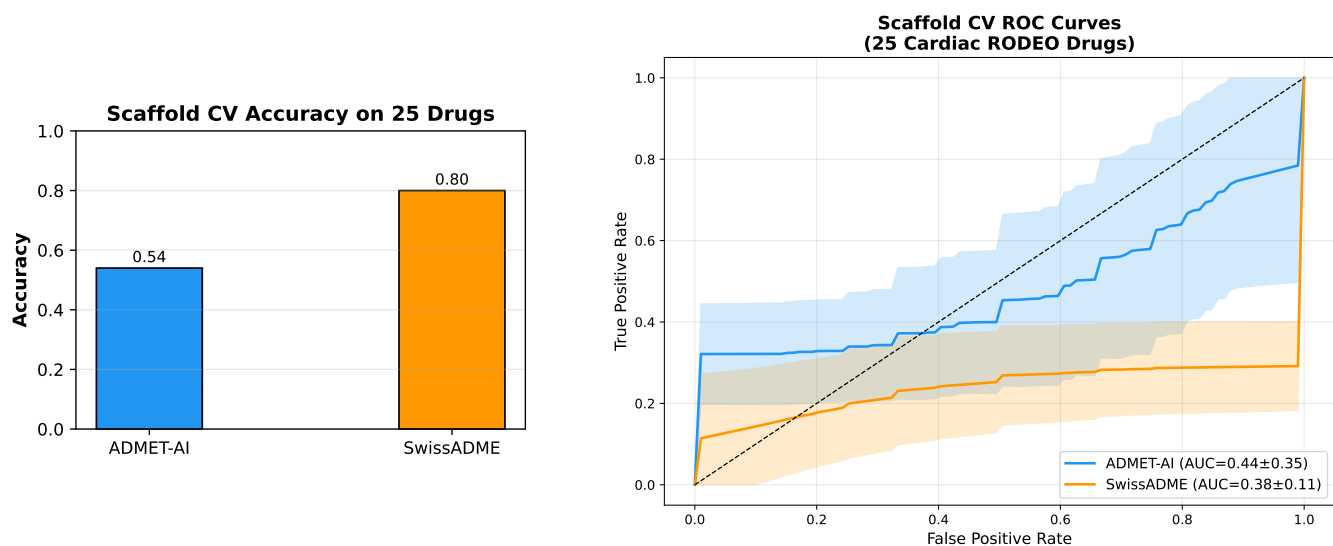


Figure 9: Scaffold CV accuracy and ROC on 25 drugs.

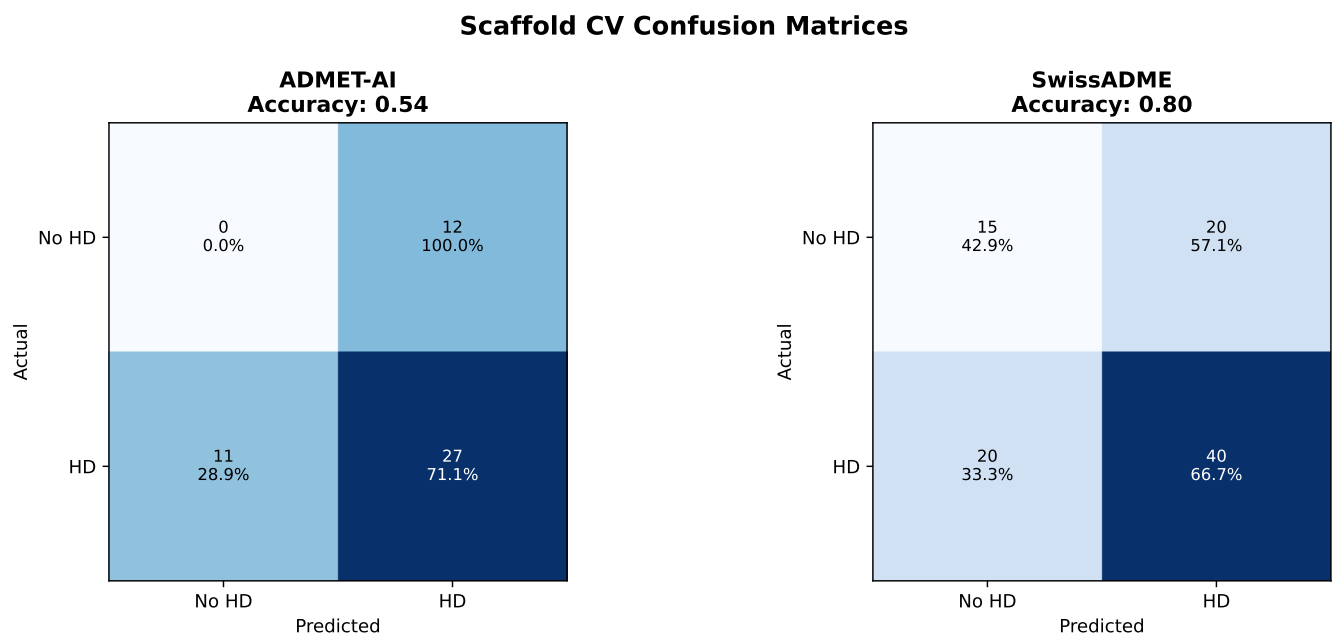


Figure 10: Scaffold CV confusion matrices.

Note: Accuracy shown in the confusion-matrix titles is fold-averaged across scaffold splits, while the matrix aggregates counts across all scaffold splits (10 seeds), so the diagonal may not match the title value.

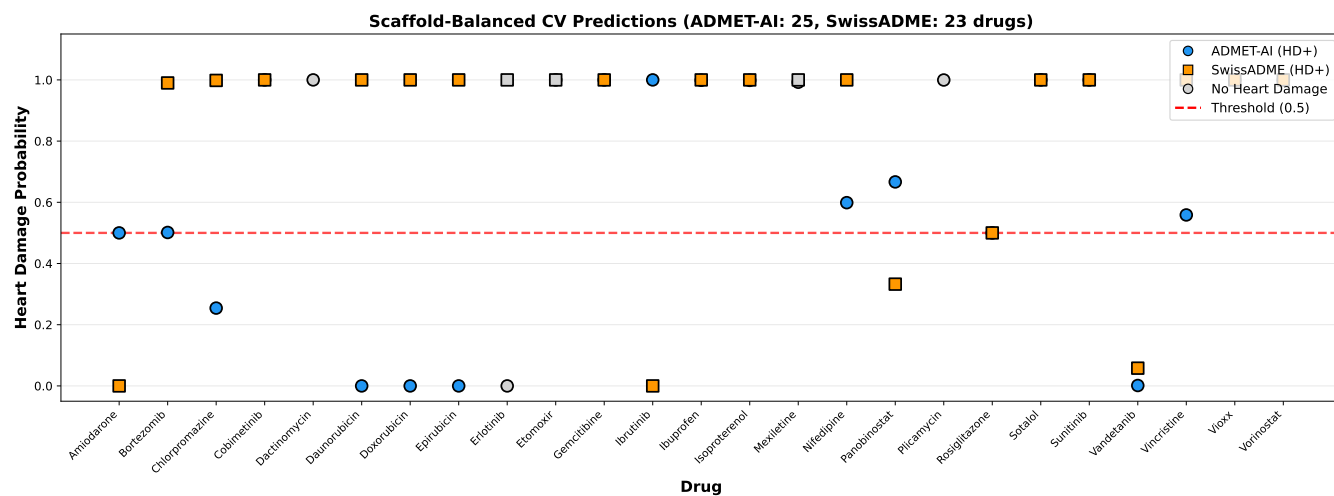


Figure 11: Scaffold CV predictions for each drug. Circles = ADMET-AI, Squares = SwissADME.

4 Organoid Model (LOOCV Comparison Pipeline)

We include the organoid-trained model from the LOOCV model comparison pipeline for direct comparison on the 25 drugs.

Table 7: Organoid Model Performance on 25 Drugs

Model	Accuracy	ROC AUC
Organoid (GaussianNB)	0.81	0.83

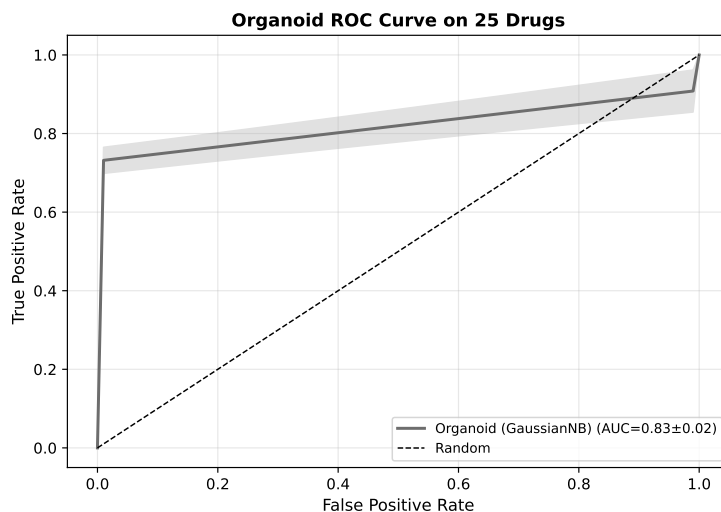
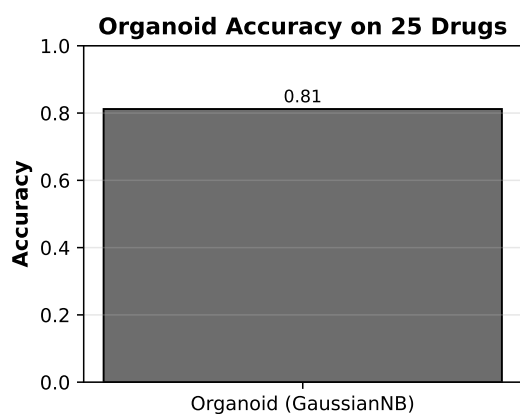


Figure 12: Organoid accuracy and ROC on 25 drugs.

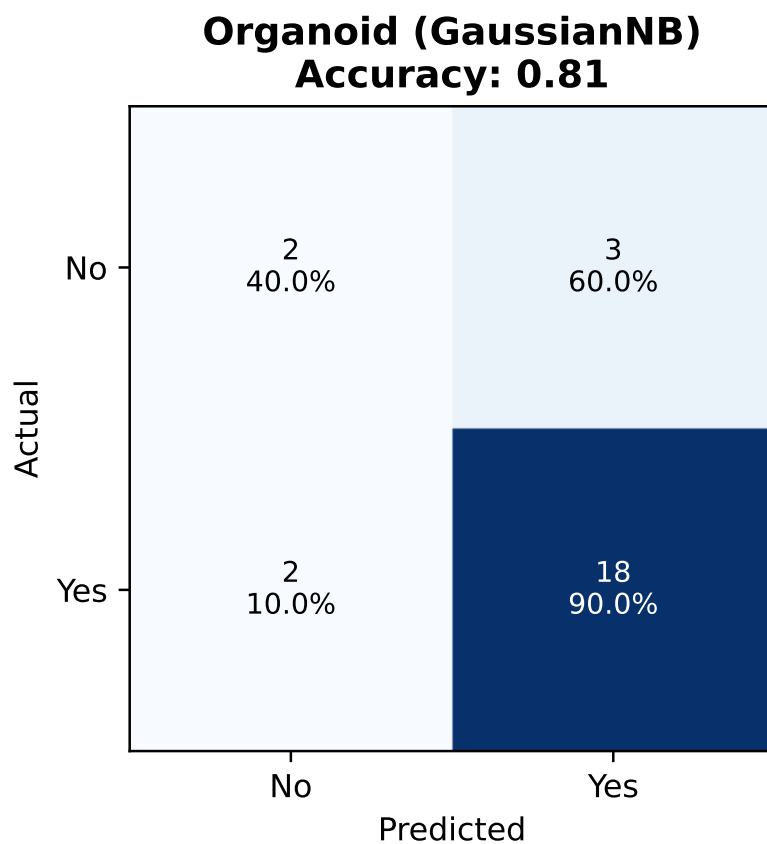


Figure 13: Organoid confusion matrix (percent correct by class).

5 Overall Comparison

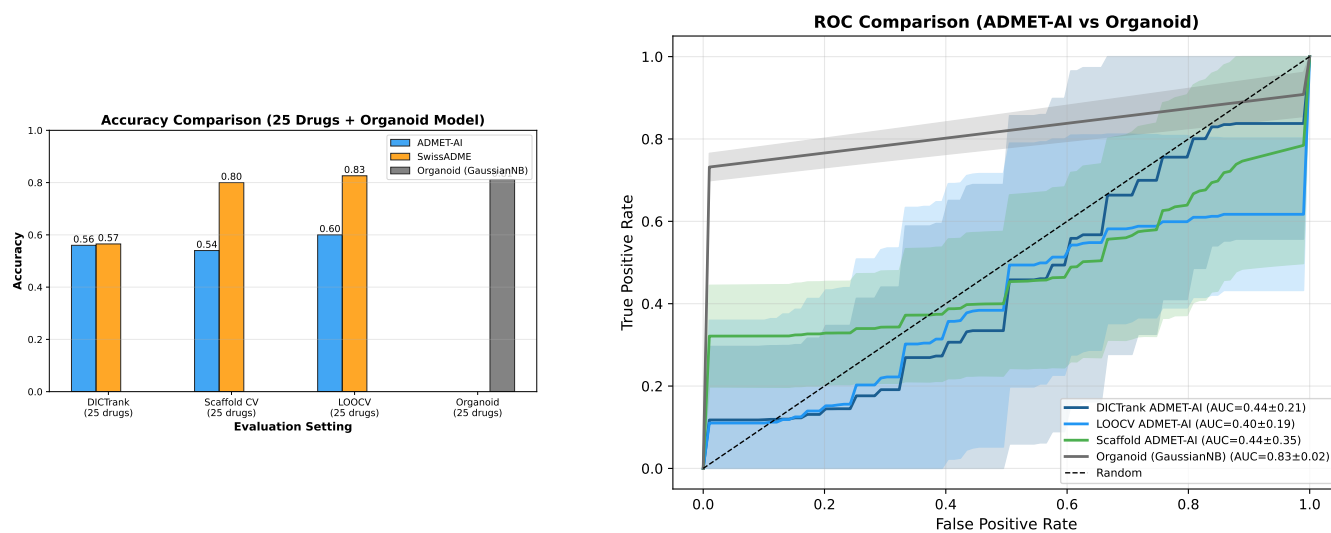


Figure 14: Overall comparison: accuracy and ROC (ADMET-AI vs organoid).

Table 8: LOOCV vs DICTrunk on 25 Drugs

Model	DICTrunk Acc	LOOCV Acc	Δ Acc	DICTrunk AUC	LOOCV AUC	Δ AUC
ADMET-AI	0.56	0.60	+0.04	0.45	0.39	-0.06
SwissADME	0.57	0.83	+0.26	0.60	0.00	-0.60

Table 9: Scaffold CV vs DICTrunk on 25 Drugs

Model	DICTrunk Acc	Scaffold Acc	Δ Acc	DICTrunk AUC	Scaffold AUC	Δ AUC
ADMET-AI	0.56	0.54	-0.02	0.45	0.44	-0.01
SwissADME	0.57	0.80	+0.23	0.60	0.38	-0.23

6 Comprehensive Performance Metrics with Standard Deviations

The following table summarizes all methods with their CV-based standard deviations. Standard deviations represent variability across cross-validation folds, providing a measure of model performance stability.

Table 10: Comprehensive Performance Metrics (Mean $\pm$ Std)

Method	Model	Accuracy	ROC AUC	F1	MCC
DICTrunk	ADMET-AI	0.56	0.45	–	–
DICTrunk	SwissADME	0.57	0.60	–	–
Scaffold	ADMET-AI	0.54 ± 0.24	0.44 ± 0.35	0.66 ± 0.26	-0.21 ± 0.32
Scaffold	SwissADME	0.80 ± 0.18	0.38 ± 0.11	0.88 ± 0.12	0.00 ± 0.00
LOOCV	ADMET-AI	0.60	0.39 ± 0.19	–	–
LOOCV	SwissADME	0.83	0.00 ± 0.00	–	–
Organoid	Organoid (GaussianNB)	0.81 ± 0.03	0.83 ± 0.02	0.89 ± 0.02	0.36 ± 0.08

Note: DICTrunk and LOOCV methods do not report F1/MCC std. Organoid std values from 5-fold CV on 25 drugs. Scaffold CV std values computed from 10-fold scaffold-balanced cross-validation.

7 Feature Importance (SHAP, ADMET-AI)

7.1 DICTrank

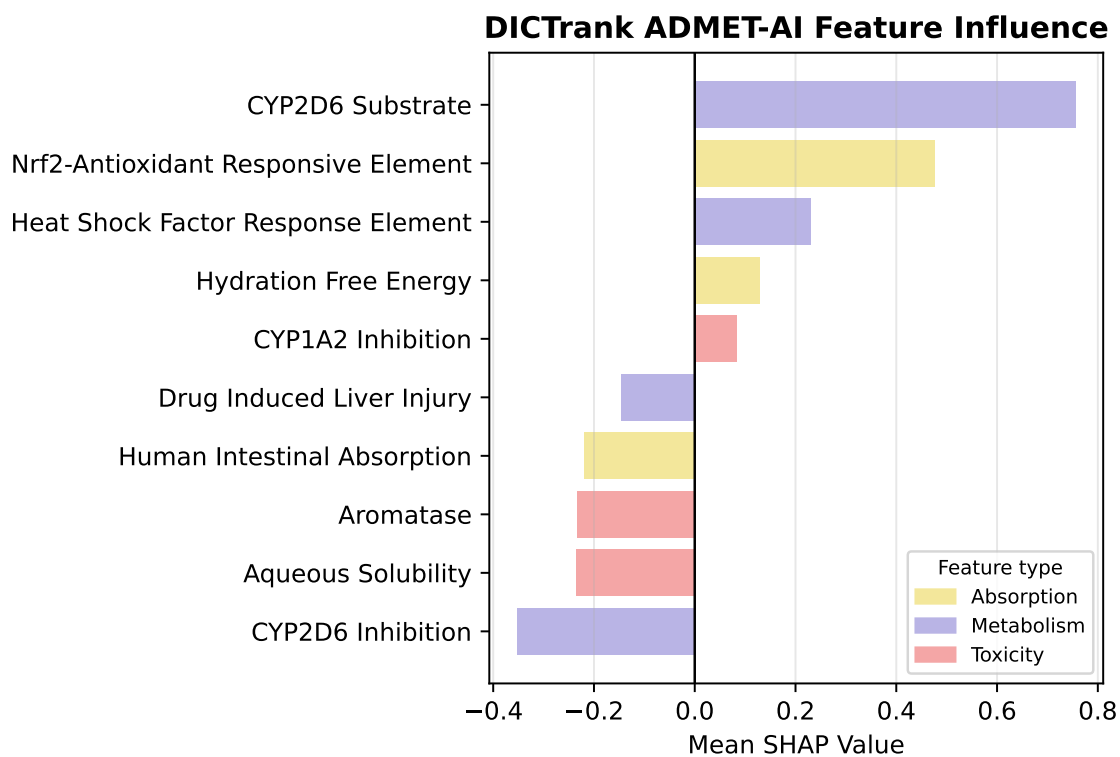


Figure 15: DICTrank ADMET-AI feature influence (paper-aligned ordering).

7.2 LOOCV

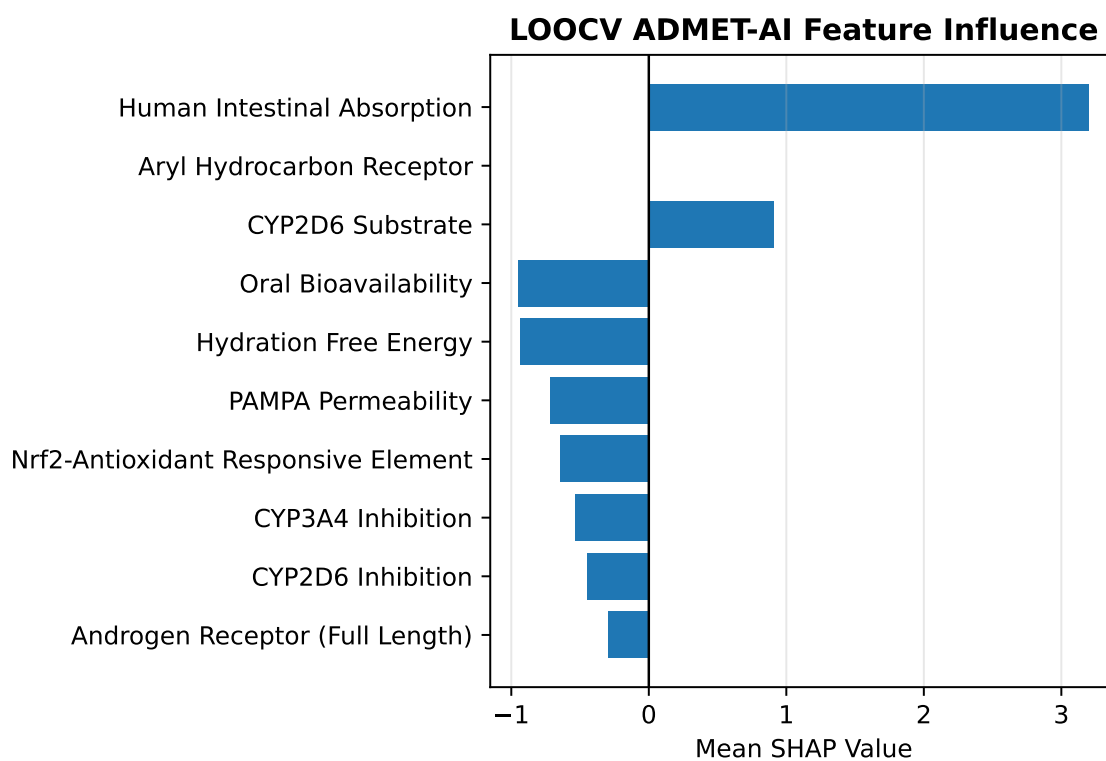


Figure 16: LOOCV ADMET-AI feature influence.

7.3 Scaffold CV

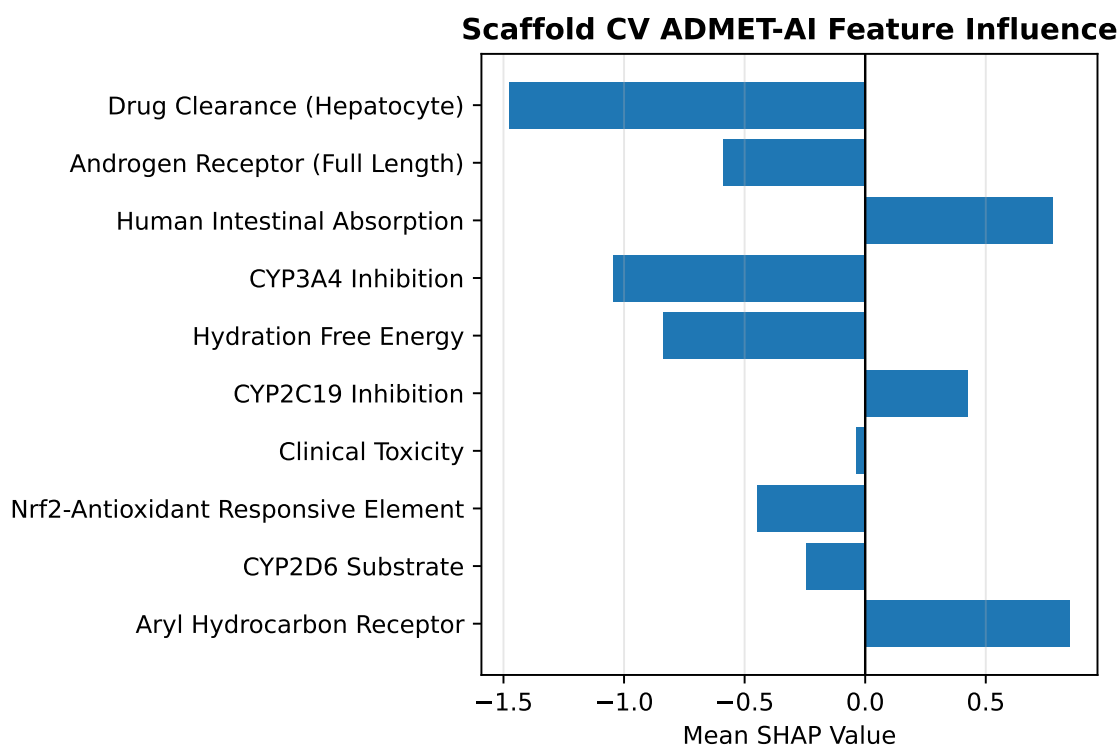


Figure 17: Scaffold CV ADMET-AI feature influence.

Supplementary: Scripts Used

- `ADMET_Comparison/Scripts/retrain_dictrank_models.py` — Retrains DICTrank models with scaffold-balanced CV and saves training metrics.
- `ADMET_Comparison/Scripts/predict_retrained_dictrank.py` — Runs trained DICTrank models on the 25-drug feature tables.
- `ADMET_Comparison/Scripts/run_predictions_v2.py` — Generates ADMET-AI features for Cardiac RODEO SMILES and DICT predictions.
- `ADMET_Comparison/Scripts/predict_swissadme.py` — Trains SwissADME model and predicts DICT concern for 25 drugs.
- `ADMET_Comparison/Scripts/full_analysis.py` — Assembles plots, tables, and the LaTeX report.
- `Prediction_Models/loocv_model_comparison.py` — LOOCV pipeline and ROC Excel export used for organoid comparisons.

References

Mukherjee P, et al. (2025). ADMET-AI Enables Interpretable Predictions of Drug-Induced Cardiotoxicity. *Clinical Pharmacology & Therapeutics*.